Vedang Deshmukh

AI / LLM Engineer · Backend Engineer (Python)

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SUMMARY

AI and backend engineer with 1 year of production experience building LLM-powered SaaS products. Sole AI engineer on a call-evaluation platform: designed and built its AWS architecture, agentic scoring pipeline (LangGraph, RAG) and compliance-minded backend. Also built a real-time interpreter dispatch system on event-driven AWS infrastructure.

EXPERIENCE

Python Developer (Backend & AI) · Arcitech.ai, Remote — Aug 2025 – Present

Intern Aug–Dec 2025, converted to full-time Dec 2025. Backend team of 4; sole engineer on the AI side.

BPOBox.ai *AI call-agent evaluation SaaS*

- Designed and implemented the AWS architecture solo (Lambda, Fargate, ECS, EC2, SQS), running evaluation workloads as serverless and containerised services.
- Built a LangChain / LangGraph agent pipeline (Claude, OpenAI; swappable to Groq or Bedrock) that scores BPO calls on tone, resolution quality and compliance using RAG (pgvector) and tool calling. Handles up to **500 calls/day**, each scored in **1–2 minutes**.
- Built a calibration module that compares AI scores with human QA reviewers, with a baseline of **87% agreement**.
- Integrated AssemblyAI for transcription and diarization. Added LLM post-processing for speaker attribution on mono audio, correcting **89 of 100** affected calls.
- Added guardrails (output schema validation, score-range checks) and LangSmith tracing to keep scoring reliable and debuggable.
- Implemented HIPAA-aligned controls for PHI/PII: transcript masking, encryption, audit logging, access control and data retention.
- Shipped FastAPI services (Django ORM) with Celery + RabbitMQ for long-running jobs and AsyncIO for higher API throughput. CI/CD on GitHub Actions, tests with pytest.

InterpretIQ *video-based interpreter evaluation system*

- Built a dispatch engine (Redis pub/sub + cache) that routes each doctor's request to the best-fit interpreter using a composite score of utilization, weighted average wait time and language availability. Matches each request in **1–2 seconds**.
- Re-architected for scalability on AWS (EventBridge, ElastiCache, SQS), replacing Redis pub/sub, Celery and Celery Beat.
- Built RBAC and user management from scratch in FastAPI: **20 default roles** plus custom roles, with JWT, OAuth and TOTP multi-factor authentication.
- Architecting a multimodal (NLP + video) evaluation engine that scores interpreter accuracy, fluency and delivery using RAG, tool calling and guardrails.

PROJECTS

Multi-Agent Research Pipeline · LangGraph, Groq, FastAPI, Streamlit

- Four agents (Planner, Researcher, Writer, Verifier) research a topic and write a cited report, with a self-correcting retry loop. Deployed on Hugging Face Spaces and Render. Code

TECHNICAL SKILLS

AI / LLM	LangChain, LangGraph, RAG (pgvector), Tool Calling, Prompt Engineering, LLM Evaluation, Guardrails, LangSmith, OpenAI, Claude, Groq, AssemblyAI
Backend	Python, FastAPI, Django, REST APIs, Celery, RabbitMQ, Redis, AsyncIO, Event-Driven Architecture, RBAC, JWT / OAuth / TOTP
Data	PostgreSQL (pgvector), MySQL, SQLAlchemy, Django ORM
Cloud & DevOps	AWS (Lambda, Fargate, ECS, EC2, SQS, EventBridge, ElastiCache), Docker, GitHub Actions, pytest, Git

EDUCATION

B.E. Computer Science (AI & ML) · Saraswati College of Engineering, Navi Mumbai — 2022 – 2026

CGPA 9.5. Student Council: Joint Head of Operations 2024–25 (events for 500+ participants, 15+ team members).

Certifications: Google Cloud skill badges (Prompt Design in Vertex AI; Gemini and Imagen), NPTEL Python for Data Science, Kaggle Intro to ML.